After running the starting file, we found that some information was hidden behind a 10,000 minute timer.
Using Cutter on the 1XL.exe file, we were able to change the assembly so it would jump to giving us the answer before the timer ran out. After running the program again, we got:
“ftp server on: 138.47.99.59”

For the server, we were able to log in with username “anonymous” and no password.
Downloaded file “telemetry” with ‘get’ command
Found:
Host: 127.0.0.1:8080
User-Agent: curl/8.5.0
Accept: */*
username: luna
password: moonrise

Icepheid
accretion
celestial
penumbra
bolide
equinox
terminator
perigee
nova
syzygy
orbit
elongation
zenith
libration
luna
retrograde
ascension
precession
conjunction
sidereal
asterism
umbra
azimuth
transit
occultation
declination
nadir
apogee
meridian
crescent
nebula
photosphere
perihelion
pulsar
parallax
lunation
solstice
eclipse
aphelion
opposition
solaris
magnitude
chromosphere
heliosphere
nutation
gibbous
albedo
ecliptic
corona
quasar

We were able to log into the FTP server and found different files using the new username and password. The names of the new directories spell out a message:

“Stars are hidden in orbit”
And at the end of the string of directories, it displays this message:
“Orbit achieved. Payload located.”

From the last directory, we downloaded the 2XL.exe file, which requires a password to get in once run in the terminal.
Using Cutter again, we changed the file to give the answer upon entering the wrong password:
“Transmission decoded. You will need this to access the archive:

```
Transmission decoded. You will need this to access the archive:
51ef2b704d359e93f4f6f50fffc53fb674c90a2c89b503b3946c561d2d9b1b4b
```

51ef2b704d359e93f4f6f50fffc53fb674c90a2c89b503b3946c561d2d9b1b4b

Using https://www.decompiler.com/jar/162f38f985a73f39bbf784a72e64fe7d/Hint.java we decoded the Hint.class file to find a hex string inside. That hex string decoded to
“hint: You are so close. You should have received a SHA-256 hash of the final password needed to open the archive. The password is a compound of exactly two words from the star catalog you retrieved earlier. You should be able to write a script to hash all possible combinations of two words from that list and compare each result to the hash you have. When you find the match, that is the password.”

Using a python script found online that utilizes hashlib and itertools, we found:
Word: nebulacorona | Hash:
51ef2b704d359e93f4f6f50fffc53fb674c90a2c89b503b3946c561d2d9b1b4b

The password “nebulacorona” is correct and we were able to get into the final pdf.
10/19/2026, 12:03PM



Contribution:
Andrew Miletello

- Found the hidden directories ““.stars | .are | .hidden | .in | .orbit” to find “-Orbit achieved. Payload located.” and retrieve all the 2””.exe files, the hidden file .message and hint.class

Hayden Mersky

- Logged into FTP server anonymously
- Downloaded telemetry file to find username and password as well as word list
- Downloaded all new files from FTP server
- Decompiled Hint.class and decoded Hex string inside to discover the hint
- Downloaded script and uploaded array of word list to discover final password

Saurav Shah

- Found the ftp credentials username-luna, pass-moonrise by analyzing the telemetry file using wireshark
- Found hidden directories one at a time using command ls -la and cd.

Amanda Berg

- Used cutter to edit the 1XL.exe file
- found the FTP server address
- Used cutter to edit the 2XL.exe file
- found the hash string of the final password
- Documentation

Ryan Bourge

- Opened the 1XW.exe file and found the 10,000 minute lock
- Used the password nebulacorona on the final pdf